1. Information economics examined through scarcity and abundance

Daphne R. Raban and Julia Włodarczyk

1. INTRODUCTION

Information economics has shown that information is a required input for and a natural output of every economic activity. A long line of Nobel laureates acknowledged information economics as a central field of study with wide ranging implications in many areas from agriculture to financial markets (Braman, 2006). Two complementary schools of thought have emerged within information economics. One school of thought investigates the fundamental assumptions of economics and shows that the availability and use of information have far-reaching effects on markets and on welfare (cf. Stigler, 1961; Akerlof, 1970; Spence, 1973; Stiglitz, 1975). The other school of thought relates to information as an economic good, an object of trade or of free sharing (cf. Benkler, 2006; Braman, 2006; Linde & Stock, 2011; Shapiro & Varian, 1999; Waldfogel, 2017). Common to both schools of thought is the recognition of inefficiency regarding information processing by markets and individuals.

In this *Companion* we posit that this inefficiency occurs both under information scarcity and abundance. The majority of prevailing models of information economics are still based on the concept of information scarcity; however, we claim that formulation of new models and theories encompassing information abundance is an important step in the development of information economics.

Information economics refers to the contribution of information to economic transactions and to the trade in information artefacts. As noted by Stiglitz (2000, p. 1471), information economics is expected to explain in detail "how and how well organizations and societies absorb new information, learn, adapt their behaviour, and even their structures; and how different economic and organizational designs affect the ability to create, transmit, absorb, and use knowledge and information". The long tradition of economic research involves analyses conducted in the context of scarcity with information discovered as a critical component of transactions.

Until the 1990s access to transaction-related information and to artefacts such as books, reports, audio-visual materials, and software was restricted and mediated. Information in its various forms was scarce and generally followed the logic of scarcity. The past 30 years have been characterized by a rapidly increasing variety of digital and networked information sources and the proliferation of both unrestricted and unmediated information artefacts. Ownership rights, which generate economic scarcity, are elusive in the context of digital information while access is a central issue. Whereas access may be restricted by techno-economic means (password, subscription), much information is freely available. In fact, information is abundantly available. We claim that abundance should be a new theoretical cornerstone of information economics to reflect reality.

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

To some extent this central theme of the *Companion* reflects Hegel's dialectics with successive phases of thesis (information scarcity), antithesis (information abundance) and synthesis. The shift towards information abundance is not a simple, quantitative increase in the volume of information available, proportional for all economic agents, nor the disappearance of the veil of ignorance. It is a qualitative change with increases in the average volume of information per agent, coupled with substantial heterogeneity in literacy levels and unequal access to abundant information. Importantly, in the real-world, scarce information coexists with abundant information exactly in the same way as inefficiencies persist, even if efficient solutions are available. This coexistence of two contrary categories (scarcity and abundance) that are simultaneously true reflects a recognition that truth involves moments that are contradictions if separated but are reconciled and consistent in their synthesis (McTaggart, 1896, p. 10). This implies that the development of economics in general, and of information economics specifically, will be driven by the synthesis of scarcity and abundance, which is far more important than simple negation. Therefore, synthesis of scarcity and abundance seems to be a non-trivial, but fascinating task for information economics.

This chapter discusses the achievements and challenges for information economics through the prism of the concepts of scarcity and abundance. First, we introduce the concept of abundance into our discourse. In the subsequent sections, we pay special attention to market operations under scarce and abundant information, as well as to prices which according to neoclassical tradition are supposed to convey all relevant market information about relative scarcity and abundance. After a short summary of conducted considerations and suggestions for future research, we present the book's structure.

2. FROM SCARCITY TO ABUNDANCE

2.1 Background

Scarcity and abundance are inherently associated with the material and immaterial (but not necessarily digital) aspects of economic activity, with scarcity often related to material processes and abundance to immaterial. On one hand, the biological construction of human beings focuses the attention of economists and policy makers on satisfaction of basic needs of a material character, while on the other hand, human cognitive development and creativity pushes people towards higher-order needs of an immaterial character.[1] Some examples of immaterial or intangible benefits include learning, self-fulfilment, satisfaction, social status, emotional well-being, and self-actualization (Swan, 2017).

In economics, the focus remains on scarcity,[2] which – confronted with limitless wants – becomes not one of the most important economic problems, but *the* economic problem of efficient allocation which theoretically can be solved via market or non-market arrangements.

Despite this strong emphasis on scarcity, abundance has started to receive increasing attention by economists (Bååth & Daoud, 2021; Dugger, 2009; Peach & Dugger, 2006; Rifkin, 2014; Sheehan, 2010; Taleb, 2012), sociologists (Abbott, 2014; Hoeschele, 2010), philosophers (Swan, 2016, 2017), legal scholars (Desai & Lemley, 2022; Madison et al., 2022), and communication scholars (Boczkowski, 2021).[3] Some attempts to conceptualize abundance echo early developments in economics. In particular, Robbins (1932) formulated two conclusions relevant for contemporary discussions on scarcity and abundance within

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

information economics. First, he insisted on the shift from defining economics as a science studying the origins of material welfare towards the emphasis on human behaviour and choice. This tendency to move away from material towards immaterial underpinnings of economics paved the way for investigations on the role of information. Second, he defined economics as a science studying the relationship between the ends of human activity and scarce means that have alternative uses (Robbins, 1932, pp. 4–15). Since then, scarcity and abundance as its opposition have been interpreted in *relative* terms.

Considering the relative perspective, abundance occurs when resources are available in larger amounts compared to wants, which may lead to negative prices (Bååth & Daoud, 2021). However, it is possible that the distribution of output and income precludes a part of society from enjoying abundance and pushes individuals into scarcity associated with poverty (Sheehan, 2010). Needs have become a moving target as marketing and advertising influence economic decisions. Moreover, marketing can be seen as an instrument of sellers to communicate a feeling of scarcity among buyers, not only in the material domain but also in the informational domain (cf. FOMO – fear of missing out). Sheehan (2010) claims that marketing is a central driver of abundance, of spending beyond the basic needs. In other words, information is a central player in the formation of general economic abundance. Spreading abundant messages (e.g., via advertising) catalyses economic transactions.

Beside this *Companion*'s focus on the dialectics of scarcity and abundance, another state between the two is sufficiency. Every agent can be assigned to one of the three systems: scarcity, sufficiency, and abundance (Daoud, 2018; Sheehan, 2010). Individuals belonging to one system experience markedly different economic conditions from people populating other systems. Importantly, the situation of individuals is determined not only by independent factors, but also their choices and attitudes. For example, self-interest appears as only one of the available options. Supported by philosophical and religious arguments people choose from the whole spectrum of attitudes: from egoistic to altruistic, from materialistic to spiritual etc.[4]

Taking into account this qualitative difference between scarcity and abundance in terms of their material and immaterial aspects, Swan (2017) interprets abundance as a social good in a broad sense including intangible qualities such as autonomy, recognition, and trust. Scarcity characterizes the material economy and describes a zero-sum game leading to competition and negative social goods. Abundance, according to Swan, is an expanding-pie model. Under abundance, digital occupations replace many material jobs. Some of the new occupations may be on the borderline of work and leisure.[5]

What brings us closer to the issue of information abundance is the observation by Dugger that "abundance is the full participation of all in the use of the community's joint stock of knowledge. And knowledge is not scarce in the sense that one person's use of it precludes anyone else's use" (Dugger, 2009, p. 21). Importantly, abundance depends on the technological advances in a society.

2.2 Defining Information Abundance

While there is no universally accepted definition, there is agreement that abundance should become more salient in economic theory. Information economics is a natural candidate to expand such discourse because regardless of socio-economic status, every user of a computer or mobile phone is exposed to abundant information and is likely to experience it with varying degrees of competence. While access and literacy may impede universal and equal usage,

Daphne R. Raban and Julia W ̶ odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

information abundance is more pervasive than other forms of economic abundance.[6] We conclude that information economics is a good starting point for understanding abundance, and in turn, abundance is a useful theoretical concept driving information economics.

As a first approximation, abundance of information can be defined as a situation when the supply of information is greater than the demand for it, even when the price of information is equal to zero.[7] Practically, we define abundance of information as

> *information exceeding informational needs or personal capacities which arrives to an agent through intuitive technologies from a multiplicity of information sources presenting high substitutability and accessibility.*

This definition requires us to clarify five issues. First, because information nowadays is mostly digital, a technological device is required for access. *Abundance depends on access*.[8] Due to the non-rival nature of digital goods, the quantity of information delivered by such technological devices is theoretically limitless. In practice, the limit is the throughput of a given technology and the cognitive ability of the user (Simon, 1955; Sims, 2003), which brings us to the second point. Information is not a regular market good. *Information is an intellectual good*. As such, it requires the user to have certain prior knowledge and competence to be able to make use of information when they are exposed to it. The feeling of scarcity or abundance of information depends on users' personal knowledge, skills, and capacities. This dependence on a reference point makes scarcity and abundance not only relative, but also subjective due to different individual evaluations (Ahituv, 1989; Kahneman & Tversky, 1979; Raban, 2007).[9] Thirdly, even though abundance may be associated with one source of information, typically it stems from a *multiplicity of sources* (cf. Simon, 1971, pp. 40–41), constituting a product of quantity of information and information sources. The fourth issue is that abundance of information can have positive or negative effects for individuals. Abundance is positive if users can derive meaning from information, for example, redundancy may be helpful to corroborate new incoming information. However, information may become superfluous, its quantity being too vast for cognitive absorption when literacy, attention, and time are scarce. Diminishing marginal returns to information may induce a point at which *abundance becomes information overload*. Finally, the fifth issue is that *beyond mere quantity, the quality of information ranges from negative (mis- or disinformation) to positive*. Even though these two issues are often interrelated, information overload and the spread of disinformation and misinformation are two distinctive aspects of information abundance.

People having to use information confront issues of access, personal ability, quantity (of information and its sources), and quality. The myriad combinations of these issues invoke different behaviours. Rational behaviour in principle consists in maximizing effects with given resources or, alternatively, minimizing resource use to achieve a given effect. Even rational agents behave differently under scarce information when they probably try to maximize effects using all available information, while under abundant information, in line with the concept of rational inattention, they would probably strive to minimize costs associated with information activity to achieve a given effect. Typically, agents are not rational, but the perspective of abundance suggests that specific inefficiencies occur as agents try to collect all relevant information to achieve the best effect under information abundance. Moreover, the character of information as a non-excludable good implies that individually rational behaviour can result in social inefficiencies which can take the form either of masses processing partial

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

information (with a lower or higher degree of heterogeneity in terms of information sets across agents) or trying to process as much of the available information as possible (also with some heterogeneity). The former may result in agents being underinformed, and the latter in agents bearing excessive costs of information activity. As argued above, both types of inefficiency can depend on access and computational power. Both kinds of inefficiencies coexist, giving rise to asymmetry of information and justifying coordination of information activity or public intervention.[10]

Summing up, the definition of information abundance relates to the relationship between the availability and variability of information itself, the users' heterogeneity, and the affordances of technology. This special relationship translates into markets in general and information markets specifically, as discussed next.

3. MARKETS UNDER SCARCE AND ABUNDANT INFORMATION

3.1 Examining Traditional Assumptions

The romantic idea proposed in the eighteenth century by Adam Smith that an invisible hand governs markets that achieve social good based on individuals' self-interest has been widely accepted and just as widely criticized. Neoclassical economists explain the notion of an invisible hand by regarding people as rational agents having access to *full information*, *efficiently* allocating scarce resources in the market to maximize their utility. This neoclassical assumption of full information ignores the issues of access, prior knowledge, cognitive capacity, and quality of information.

Profound criticism of the assumption of acting upon full information came from realizing that in real markets, availability of *information is asymmetric* (Akerlof, 1970). Asymmetry of information occurs for both sides of the market creating relative scarcity on one side of the market. Producers and sellers (supply side) usually possess more information than buyers about the quality of goods. Buyers (demand side) possess more information about their own preferences, socio-economic status, or health. Information asymmetry leads to various forms of market failure including adverse selection, moral hazard, excessive market power (monopolies) and public goods. Information economics discussed strategies, specifically signalling and screening, which are meant to mitigate asymmetry, reduce uncertainty, and enable better functioning of markets (Spence, 1973; Stiglitz, 1975).[11] These approaches have special merit when scarcity is concerned. Market failure under abundance has yet to be modelled to suggest novel solutions, and, where needed, external regulation.

Approximately in parallel to the realization of the importance of information for the existence and functioning of markets, development of behavioural economics has shown that the other two assumptions, rational choice and utility maximization, do not reflect the behaviour of the majority of people (Kahneman & Tversky, 1979). Through experimentation, psychologists and economists have shown that people are *inefficient* in their information usage which leads to suboptimal decisions and markets providing goods and services to irrational actors (Bastardi & Shafir, 1998; Shampanier et al., 2007; Thaler, 1991; Tversky & Kahneman, 1982).

As mentioned above, markets operate in a continuum of scarcity – sufficiency – abundance. Theoretically, only the situation of sufficiency is informationally neutral, generating no

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

informational frictions impinging on market efficiency. However, as pointed out earlier, in practice it is unlikely that *all* economic agents would have similar access and competences to process this sufficient information. Moreover, the easy availability of information at no cost pushes people towards overconsumption making the state of abundance the default or the new reference point.

3.2 Information in Markets

One of the most important questions is: how does the abundance of information change the operations of markets? To answer this question, we need to realize that markets are essentially about exchanging, transforming, and monetizing information. Information is a central source of value in many material markets as manifested by the major contribution of elements such as product development, design, branding, and customer service. Markets create incentives for an exponential growth of information production, as well as for the spread of disinformation and misinformation. Abundance of information can be thus perceived as a result of market changes (concentration) and a catalyst of further changes. In other words, information is an element of a powerful feedback loop pertaining to structural changes in many markets.

Exponential growth of information production requires access to advanced technologies and hardware which induces market concentration in the form of platformization in the digital world. These technologies, together with data and information constitute barriers to entry to the oligopolistic or even monopolistic market of information. Other markets become subordinate to the information markets and their operations are often limited by constraints imposed by information giants and their algorithms. When algorithms aim to monetize popularity instead of truth, the invisible hand of the market becomes the 'wisdom' of crowds, resulting in an ever-increasing market concentration.

Platforms thrive on production of data and information; however, they tend to have little interest in the content (Ma, 2023), as long as it has no direct impact on the platform's profits. Often the quantity rather than quality of information on platforms drives profits. Dominant platforms determine what kinds of information can be disseminated. Their concentration on profits involves actions appealing to a large audience, which effectively means an agreement on deviations from the truth. Under platformization, the informativeness of information is not as important as monetization potential. As a result, platforms have minimal interest in distinguishing between information of high or low quality, or between facts and fake news. Furthermore, even if popularity can reflect informativeness, popularity can be also shaped by algorithms (Ma, 2023).

There are also subtle, personal and social, mechanisms impinging on the operations of markets under the condition of abundance. For instance, Gino and Pierce (2009) experimentally find that the context of monetary abundance induces envy and significantly changes people's behaviour to increase cases of cheating. Even though their research focused on the abundance of wealth, not information, we observe the occurrence of behavioural effects under information abundance as well, e.g. attitudes towards vaccination.

This picture becomes more complicated, because even under abundant information some agents operate in a relatively scarce environment compared to others. Agents are heterogeneous in their preferences, prior knowledge, cognitive capacity, access to digital resources, and literacy levels, all of which may generate relative scarcity, especially as they pertain to the use of information as market enabler in the exchange of various goods and services. This agents'

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

heterogeneity may be reinforced by phenomena such as rational inattention or information self-rationing. Information abundance in this case can be instrumental to achieve potentially more efficient allocation of resources (with or without direct impact on prices of the goods involved).

3.3 Markets of Information

In the case of the markets for digital information goods, abundance and agents' heterogeneity have an impact on market structure and prices of information goods. Such markets are especially susceptible to market failure. In fact, market failure has become *de facto* a business goal. Operating within the scale-free network structure, entrepreneurs understand that their strategic goal is to be the 'winner takes all' of the network in their selected area of activity. Monopolizing a networked market means owning the market activity data. Application of advanced artificial intelligence algorithms on owned data bolsters the monopolistic position. The proliferation of network-based monopolies means that instead of a free information market, centrally controlled information silos are the current aspiration and position. Financial, economic, and political power ensue. Recent technological developments present blockchain as a solution based on the idea of decentralization. While some perils of cryptocurrencies based on blockchain technology are evident, we assume that solutions will develop to stabilize currency markets. Blockchain evangelists envision a world where commercial, business, and managerial transactions are conducted at the users' free will. This being an ever more virtual environment, users will need even more advanced digital skills to manage their digital identities as part of virtual life.

While the early decades of internet growth have given rise to online communities in parallel to business and commerce, current developments such as blockchain technology and the vision of a metaverse focus on markets and monetization. Blockchain technology marks a major step towards a pure market economy because it is predicated on assuring scarcity and ownership rights. To what extent can consumption shift from purely material and wasteful to purely informational, partially decreasing the pressure on natural resources (with a significant exception concerning energy sources)? Is this an innovative opportunity for information economics to reduce material consumption rather than induce it? For example, buying non-fungible tokens (NFTs) in the form of pixel art can provide a modern form of conspicuous consumption. NFTs, which are based on blockchain technology, are an instrument for creating scarcity and maintaining ownership rights for digital artefacts. The opacity relating to blockchain developments and the inclination of information markets to monopolization raise an urgent need to investigate mechanism design in the presence of abundant information products.[12]

An example of the effect of abundance on the operation of information markets is salient in everyday life where we observe the entanglement between excessive quantity and inferior quality of information. Due to broad access, existing knowledge and beliefs, limited literacy, and zero marginal costs, we observe the ever-increasing spread of mis- and disinformation, also known as fake information, and the proliferation of spam.[13] When abundant information is not necessarily reliable, metainformation (complementary information) may be needed – greater access to information thus stimulates greater demand for information. This demand for metainformation is also driven by the nature of information as an experience good. This entails economic dynamics where utility becomes known *ex post*, but the decision to engage in information activity has to be made *ex ante.*

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

3.4 Information Market Dynamics

Dugast and Foucault (2018) introduce dynamism into the analysis of abundant information. They explain that it takes time to analyse abundant data and text (tweets, company reports and more), therefore, inevitably raw information that arrives quickly is of lower quality than analysed information that takes a longer time to produce. Noise comes before the signal.

The decrease in the cost of producing low precision signals due to information abundance reduces the profitability of trading on subsequent (more precise) signals and crowds out incentives for fundamental research (Dugast & Foucault, 2018). Thus, information abundance can reduce the long run price informativeness. As a result, the noise-based behaviour of agents can change the market reality, so that the behaviour of better-informed agents can potentially become irrelevant to the new market situation.

More precisely, the mere presence of information delay may not be sufficient to explain potential market outcomes under information abundance, especially when it is framed within the information-delay variant of the rational expectations theory, relating behaviour to optimally formed forecasts of the future. Allowing agents to act upon inferior forecasts (e.g. due to information processing constraints) brings us closer to the rational inattention theory proposed by Sims (2003). Rational inattention theory considers systematic changes in the nature of noise due to dynamic properties of economic processes. These changes explain *inter alia* reallocation of information monitoring capacities towards areas characterized by greater noise associated, for example, with a greater variability of prices (Sims, 2003), because such areas offer potentially higher profits. Importantly, deviation from optimal behaviour predicted by rational expectations theory introduces heterogeneity, increasing the complexity of market phenomena pertaining to information processing.

3.5 Abundance of Data

Our focus is on information; however, an emerging literature on abundance of data, which is the raw material of information, deserves mentioning. For instance, Acemoglu et al. (2019) explain why data are given away freely by users to platforms which use data to extract high profits: when a small group of users is willing to share data as part of their participation in a network that provides insight about other users, by inference. Even if those other users regard their data as valuable, they cannot form a market because their private data have already been inferred. This is a specific form of adverse selection. Acemoglu et al. (2019) do not provide reasoning for why the first users are willing to give away their private data. The answer probably lies in the value they perceive from being active in the network they selected (Benkler, 2006; Braman, 2006).

When firms own abundant data, they may overuse it and not adequately respect consumer privacy. Data are non-rival enabling infinite and simultaneous reuse. Yet, data are often difficult to copy and can be made excludable, so a market is possible, creating even more value for firms. Companies have the means to aggregate and analyse data to produce information, an ability that individual lay users currently lack.

While privacy remains an important concern,[14] Acemoglu et al. (2019) and Jones and Tonetti (2020) demonstrate some benefits of data abundance such as the ability to draw inferences, generate innovation, reuse resources, make better decisions, offer recommendations, create new knowledge and more. However, these benefits are asymmetric. Data held back by

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

companies reduce social welfare. Some researchers suggest that private ownership of data will improve social welfare through the development of efficient data markets (Jones & Tonetti, 2020; Mansour et al., 2016). Yet, given what we know about information economics and agents' heterogeneity, there is no reason to believe that individuals will exercise their market opportunities efficiently. There may be a need for the government to create or regulate incentives that would shape this new data market.

In section 3 we outlined how information abundance influences markets and suggested that markets should be designed under the assumption of abundance of information and information goods. Within such a design, prices play an important role when they are present but also when they are absent. While markets rely on the informativeness of prices, information itself often appears in markets for free. Priceless information can be price-free, which we unpack next.

4. PRICES AND BEYOND

In Oscar Wilde's famous play *Lady Windermere's Fan*, Lord Darlington explains to Cecil Graham that a cynic "is a man who knows the price of everything and the value of nothing". Cecil Graham replies: "And a sentimentalist, my dear Darlington, is a man who sees an absurd value in everything, and doesn't know the market price of any single thing" (Wilde, 1893, pp. 95–96). This literary conversation is a fairly good representation of the current state of economic understanding of prices and value.

When scarcity is the frame of reference, prices convey some information about wants and preferences, as predicted by Hayek (1945). However, the informativeness of prices is limited because it does not account for information asymmetry nor for the subjective perception of value. If prices had conveyed all the necessary information at no cost, markets would be frictionless and efficient, yet, more than half a century of research tells us that markets suffer from informational inefficiency (cf. Akerlof, 1970; Grossman & Stiglitz, 1980). Moreover, in a digital world where abundance of information is the frame of reference, we observe a new reality where zero prices are prevalent. In other words, information abundance is associated (but not necessarily causally related) with the price of zero.

Information is known to have high costs of production and virtually zero marginal costs. These traits coupled with non-rivalry and non-excludability attributes often lead to the conclusion that a price of zero is inevitable. The result is overconsumption of free information (Dugast & Foucault, 2018; Raban et al., 2019; Raban & Koren, 2019) to the point that it becomes harmful as evidenced by the recently coined term, infodemic (Eysenbach, 2020; Global Infectious Hazard Preparedness, 2021). Is it an efficient market? Observing various phenomena such as deceitful information (e.g. fake news, urban legends, spam) and information overload casts serious doubt regarding information market efficiency. While mis- and disinformation have been practised by leaders and institutions for thousands of years, also under information scarcity (cf. Machiavelli, 1532; Tzu, 2010), nowadays they may be initiated and promoted by anyone. One of the important problems refers to information of poor quality crowding out information of good quality, because at the end of the day we can conclude that even if information in general is abundant, information of good quality becomes scarce.

Research has shown that zero prices increase consumer demand regardless of need or quality (Shampanier et al., 2007). Exposing people to an economic framing (i.e. requesting

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

payment for something that was previously free) also results in a change of preference such as more focus on utility or usefulness (Heyman & Ariely, 2004; Raban et al., 2019). In other words, zero prices invite information overload while economic framing induces an evaluative mindset. Could it be that information markets are missing price informativeness when prices are absent? Whereas the absence of a price may carry implicit informativeness, we suggest that digital implementation of price informativeness such as gamified prices could be an avenue for introducing reliability while affecting users' epistemic approach and without impeding access.

The complementary aspect of prices, which received considerable attention via the prospect theory, is the notion of value indicating individual differences in the perceptions of prices and risk (Kahneman & Tversky, 1979). Subjective perceptions of value are part of the information not necessarily conveyed by prices (Raban & Ahituv, 2024; Raban & Rafaeli, 2006). For example, people may act emotionally while making economic decisions. Such emotions are triggered by the value system, not by the prices on the market. The perceived value system comes into play in participatory pricing schemes such as in bargaining or auctions. Auctions have been successfully implemented in Google's advertising pricing; however, participatory pricing can be further developed in many other digital information markets. Similarly, voluntary pricing and payment such as pay-what-you-want or name-your-own-price have remained largely esoteric options in information markets despite their unique suitability for experience goods. It could be that companies, and especially monopolies, prefer to exercise price differentiation through big data analytics. Relying on big data analytics may be likened to driving while looking at the rear-view mirror. We suggest looking through the windshield to identify new behaviours and detect new potential via value-based participatory or voluntary pricing and payment. Through systematic discussion of payment schemes for information products, Stock (2024) exemplifies that prices are not merely supplier-provided numerical indications of market activity. Prices as carriers of value can be numeric but may also appear in terms of time, attention, reputation and more.

While scarcity in information markets leads to higher prices and lower access, abundance does not guarantee universal access (cf. Plamondon, 2022). Abundant information does not even reduce price dispersion (Goldfarb & Tucker, 2017). In fact, the online environment with its abundant data and personalization options is conducive to price discrimination tactics and constitutes a significant opportunity for capturing consumer surplus.

Demand for information products can be seen by vibrant search activity, news consumption, app downloads, book sales and a host of other indicators. Some of this demand activity is market based, albeit information markets tend to be monopolized (Levitan, 1982; Linde & Stock, 2011; Shapiro & Varian, 1999). Much of the demand for information addresses free sources, which is to be expected. While markets for information products exist, there is vast opportunity to innovate through pricing with money or non-pecuniary currencies such as reputation, attention, influence and others which are often complementary and non-rival and are part of the mark of abundance.

5. SUMMARY AND IDEAS FOR FUTURE RESEARCH

The neoclassical approach, according to which information is a public good defined by what it is not – non-rival and non-excludable – has limited relevance to analyse abundant information. This chapter is an attempt to come up with new, positively framed attributes which can be used

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

to describe information markets. Information is an intellectual good which is often abundant, has variable value, requires skills and is frequently free of direct charge. Novel digital accounting mechanisms are being continuously developed and implemented to assist in filtering and verifying abundant information.

The tools of economics enabled important insights regarding asymmetry and market structure, and the resulting market mechanisms and business models. In this regard it is striking to realize that economic analysis has largely overlooked the intriguing phenomenon of information abundance. While there have been attempts by economists to quantify this phenomenon (Lyman & Varian, 2000), these attempts were soon neglected due to the rapid quantitative increase and the understanding that the significance of information is beyond quantity. The qualitative change occurring in the switch to an abundance frame of thought is multi-dimensional, with quantity being one of various aspects to analyse. Additional dimensions appeared in our definition of information abundance, and it is likely that more dimensions will arise with added research attention. What is the meaning of information abundance for any kind of market? What is its particular meaning for information markets? We have started to lay the ground for incorporating abundance into economic analyses; however, these questions await theoretical and empirical economic investigation.

It may seem that if scarcity of information to consumers creates an informational problem, then abundance of information may be a solution. In fact, information abundance is not an efficient remedy for problems prevailing under information scarcity. Abundance of information does not imply the end of information imperfections. It produces new imperfections which are still not well characterized from an economic perspective.

The effects of abundant information may be studied at the individual level, at the market level and from a regulatory perspective. At the individual level, access to information, prior knowledge, and literacy interact with informational variables such as price of information (zero or non-zero) and information quality in ways which require theoretical and empirical scrutiny. The heterogeneity of potential strategies associated with dealing with abundance of information (e.g. due to rational inattention) matters for market efficiency.

At the market level, we observe two general types of market: markets for non-information goods and services and the market for information artefacts. Both market types suffer from a host of market failures necessitating the conceptualization of specialized innovative mechanism design that would incorporate the characteristics of abundance. We suggest studying how to include the informativeness of prices even into exchanges of free information to gain the positive effect of price information. The issue of obtaining meaning from information such as in learning or expert advice or analytics is also a good candidate for market development. Additional innovation could come from envisioning virtual marketplaces that apply reliability and verifiability as part of their structure. The notion of non-pecuniary payments such as likes or kudos give rise to specific 'banking systems' for such assets (sometimes in a positive sense, sometimes creating speculative and fraudulent activities including click farms). These ideas await theoretical and empirical support.

The tension between information scarcity and abundance, especially the attempts of private entities to monopolize data and information to exploit and exclude worse-informed parties constitutes a rationale for public intervention. This is where policy research can develop while taking maximum care to preserve the Jeffersonian spirit of the free spread of ideas (Jefferson, 1905). Through regulation, market transparency may improve and market failure, which is always anticipated for information, could be avoided in advance, by design (Stiglitz, 2009). In

Daphne R. Raban and Julia W ̶ odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

fact, there is also uncertainty about scarcity and abundance of information in the future. While the volume of information may increase, the proportion of information publicly available to the total volume of information may decrease over time, especially when dominant platforms will try to increase the costs of changing the standard by closing consumers and other agents in their networks (lock-in) (cf. Włodarczyk, 2020, p. 19).

The science of economics has been transformed with the introduction of psychological research as indicated by the 2002 Nobel prize for behavioural and experimental economics. The current challenges in information economics, specifically the implications of abundance, invite a collaboration between economists and researchers from the fields of information science, learning, digital literacy, information systems, computer science, psychology, sociology and more.

6. ABOUT THE CONTENT OF THE *COMPANION*

This *Companion* is an edited volume dedicated to information economics written for an audience of scholars and advanced students. The chapters are grouped into seven sections, each touching on a significant area of the field as a whole. While overviewing the chapters, some tensions and issues become visible. The longstanding tension between free markets and government regulation is most pronounced in Chapters 2, 8, 13, 23, 24, and 25 with some relation also in Chapters 3 and 16. Tension arises from the often-extreme positive network externalities when confronted with local compromises such as individual privacy or effort. When positive externalities serve the public good, then a compromise on individual concessions may be acceptable; however, due to market failure the externalities often fortify monopolistic power. This tension appears, for example, in Chapters 21 and 23. Chapters 12, 20, 21, 23, and 24 provide some details relating to issues of particular affordances of information technologies that are not always known or understood by users, which imply that technology itself creates asymmetry. Issues that relate to problems with distribution of information and resources which eventually lead to inequality are highlighted in Chapters 2, 4, and 20.

The tension between scarcity and abundance is central in this volume. In Part I, entitled 'Information Economics Overview', Chapter 2 reviews information economics research which is premised on scarcity highlighting the critical role of information in the failure of the fundamental theorems of welfare economics, the non-existence of competitive equilibrium, and the dependence of the nature of the equilibrium, when it exists, on both what information is available, and how information can be acquired. Chapter 3 acknowledges the development of information abundance, its current manifestations, harmful effects and methods to control it. Chapter 4 goes on to show how scarcity and abundance of information play out for income distribution.

Part II deals with 'Information Asymmetry', which occurs when one or both sides of a transaction experiences relative information scarcity. Chapter 5 offers a historical perspective on the breakthrough associated with inclusion of asymmetric information in the discourse on market failures, while Chapter 6 attempts to summarize the abundant scientific research in the area of information asymmetry (1979–2021) using bibliometric analysis. Chapter 7 claims that advancement of data collection and artificial intelligence shifts the problem of asymmetry of information and that theoretical models should be replaced or supplemented by data-driven approaches such as machine learning, provided that monopolies of knowledge do not emerge.

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

Chapter 8 links asymmetric information to specific aspects of private information on the healthcare market (health insurance, doctors' prescription and referral strategies), and studies the effects of contract regulation of the healthcare market. Chapter 9 discusses consequences of information scarcity and abundance epitomized by uneven social embeddedness of venture capital funds in Israel, operating under conditions of severe uncertainty, asymmetric information, and high financial risks. It emphasizes the role of network position determining *inter alia* accessibility to privileged information flows.

Part III is about 'Information Transmission and Influence'. In a sense, this section is about information sufficiency in the context of decision making – finding ways to convey the appropriate information in decision tasks. Chapter 10 reviews and expands the theoretical literature on disclosure of conflicts of interest in expert/decision maker relationships. Chapter 11 discusses four aspects of strategic information transmission, such as the extent to which delegation can be superior to communication, the role of experts' competition, the effectiveness of reputation, and the role played by networks. Chapter 12 discusses social influence in e-commerce in the form of the impact of other consumers affecting the decision making of buyers and demonstrates how a machine-learning model can forecast future ratings and reviews.

Part IV is about 'Innovation and Intellectual Property'. Innovation increases information abundance. It can be motivated by information scarcity, but is usually enabled by its abundance (especially in case of combinatorial innovation). Intellectual property rights are means for implementing scarcity when innovation is concerned. Chapter 13 develops an integrative approach to innovation and focuses on digital innovation, which raises unique challenges for economic theory due to its plasticity and open-endedness. Chapter 14 investigates the relationship between information and innovation in an evolutionary-institutional perspective (demonstrating that complexity can be prohibitive or largely counterproductive to innovation), interprets the basic model of 'evolution of cooperation' in terms of information and innovation implications, and considers the significance of network structures and of 'self-organization' mechanisms. Chapter 15 makes an attempt to systematize the main concepts and approaches to intangibles, information goods, and intellectual property goods which can be described as hybrid goods made up of information and law. Chapter 16 acknowledges that the current implementation of intellectual property rights is not well-suited to knowledge innovation and suggests alternative institutional configurations to accommodate knowledge building processes.

Part V on 'Payment, Value, Crowdfunding' highlights individual differences and social processes that underlie information monetization. Chapter 17 provides a comprehensive overview of payment methods including direct, indirect, and social forms of monetization. Chapter 18 describes the theoretical basis and empirical evidence regarding perception of value of information goods and outlines the significance of value perception for decision making and for construction of markets. Abundance of information highlights the need to polish digital literacy in addition to economic choices. Willingness-to-pay scales appear in Chapter 18 and in Chapter 19 which is devoted to the special case of knowledge influencers, who may be viewed as gatekeepers in the presence of excess information. Following theory development, the chapter examines how the attractiveness of influencers increases attachment to and willingness-to-pay for knowledge products. Chapter 20 raises social considerations of crowdfunding for cultural products by focusing on social and cultural capital and by explaining sources of inequality in raising funding.

Daphne R. Raban and Julia W ̶ odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

Part VI is devoted to 'Challenges'. While earlier challenges such as information asymmetry were associated mainly with information scarcity, current challenges are often about information and data abundance. Chapter 21 problematizes bulk organizational email and the associated costs. It then suggests a multi-stakeholder economic model to alleviate the problem. Describing an empirical study, Chapter 22 highlights the importance of supervision for maintaining quality of information when engaging with accounting information systems. Chapter 23 illuminates some dark corners of technological progress, namely, using advanced artificial intelligence algorithms to process large amounts of user data results in potential breaches of privacy. Technological knowledge is needed to manage these threats and devise appropriate policy. In closing the challenges section, Chapter 24 reminds our readers that alongside the technological developments (internet of things, artificial intelligence, nano-devices, virtual and augmented reality, 3-D printing and many others) and the associated information and data, the digital environment holds many opportunities for illegal activities.

Finally, Chapter 25 in Part VII, 'Future', draws together the main threads discussed in this volume and presents cross-disciplinary approaches to information economics, starting from the historical background and moving towards the implications of artificial intelligence. The chapter discusses foundational terms that would enable information economists from different subfields to understand differences in their research agendas and methods. These terms require changing from thinking about economic agents to perceptual entities; from treating information as data points to differentiated information; and from focusing on markets to looking at probability spaces.

NOTES

1. This perspective corresponds to Maslow's hierarchy of needs explaining differentiated motivation under scarcity (satisfaction of basic needs reduces motivation) and under abundance (motivation increases when higher-order needs are satisfied) (Maslow, 1943). This has important implications for economies, especially capitalistic ones, which are based on scarcity.
2. A doctrine particularly explicit about scarcity was formulated by Thomas Malthus (1798) who treated poverty and famine (evident manifestations of scarcity) as natural outcomes of fast population growth relative to resources. Interestingly, Malthus acknowledged that intellectual needs are not as satiable as others and can be treated as higher-order needs: "When the mind has been awakened into activity by the passions, and the wants of the body, intellectual wants arise; and the desire of knowledge, and the impatience under ignorance, form a new and important class of excitements" (Malthus, 1798, p. 377). However, he also noted that "Could we suppose the period arrived, when there was not further hope of future discoveries, and the only employment of mind was to acquire pre-existing knowledge, without any efforts to form new and original combinations, though the mass of human knowledge were a thousand times greater than it is at present, yet it is evident that one of the noblest stimulants to mental exertion would have ceased" (Malthus, 1798, p. 383).
3. Earlier works include e.g. Ward (1966) and Fox (1967).
4. See for example Stoic philosophy and the recommendation to equate sufficiency with abundance: "Whatever is enough is abundant in the eyes of virtue" (Seneca, 1932). Drawing from Buddhist and Islamic readings, Rafikov and Akhmetova (2019), like Seneca, suggest simplicity and spirituality as a solution for both waste and scarcity, so that more people would feel relative abundance and contentment.
5. Chapter 4 of this *Companion* alludes to the concept of playbour (Włodarczyk, 2024, pp. 81–104).
6. Beyond information economics, abundance is typical also in 3D printing, synthetic biology (potentially revolutionizing production of food and medicines), robotics (Lemley, 2016) and more.

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

7. From the suppliers' perspective, this zero price of information may translate into losses as the supplier incurs costs they cannot recover directly from the consumers.
8. Abundance of a good or service itself does not guarantee a universal access to it and may give rise to inequalities (cf. Plamondon, 2022; Włodarczyk, 2024).
9. Interestingly, the prospect theory with its assertion that losses are felt more intensively than gains, implies that scarcity of information is experienced more intensively than abundance (which also justifies the focus on scarcity prevailing in the literature). Furthermore, when abundance transforms itself into information overload, it becomes a burning problem again.
10. The alternative options stemming from the principle of rational behaviour can be linked to different motivation of individuals. Some of them may be motivated by aversion towards waste (e.g., want to rent their temporarily unused premises), while others remain profit-oriented (don't mind having unused premises if they can rent them at a higher price). The former is more inclusive, the latter more exclusive, but both are stimulated by greater availability of information about capacity use etc.
11. Chapter 2 in this *Companion* provides a detailed historical account and current knowledge on information asymmetry (Stiglitz & Kosenko, 2024a, pp. 20–52).
12. Chapter 17 of this *Companion* describes information prices and payment for information goods (Stock, 2024, pp. 339–363), while Chapter 18 applies the approaches of behavioural economics to the study of information as a good offered on the market (Raban & Ahituv, 2024, pp. 364–378).
13. Chapter 3 of this *Companion* relates to mis- and disinformation (Stiglitz & Kosenko, 2024b, pp. 53–80) and Chapter 21 relates to spam (Konstan & Kong, 2024, pp. 407–435).
14. Chapter 23 of this *Companion* details how technological advances present ever-growing challenges to privacy (Bodoff, 2024, pp. 462–480).

REFERENCES

Abbott, A. (2014). The Problem of Excess. *Sociological Theory*, 32(1), 1–26.

Acemoglu, D., Makhdoumi, A., Malekian, A., & Ozdaglar, A. (2019). *Too Much Data: Prices and Inefficiencies in Data Markets*. National Bureau of Economic Research, No. w26296. https://economics.harvard.edu/files/economics/files/acemoglu_spring_2020.pdf.

Ahituv, N. (1989). Assessing the Value of Information: Problems and Approaches. In J. I. DeGross, J. C. Henderson, & B. R. Konsynski (Eds.), *Proceedings of the Tenth International Conference on Information Systems* (pp. 314–325). https://aisel.aisnet.org/cgi/viewcontent.cgi?article=1007&context=icis1989.

Akerlof, G. A. (1970). The Market for "Lemons": Quality Uncertainty and the Market Mechanism. *The Quarterly Journal of Economics*, 84(3), 488–500.

Bååth, J. & Daoud, A. (2021). Extending Social Resource Exchange to Events of Abundance and Sufficiency. In *Dictionary of Ecological Economics* (pp. 2015–2017). http://arxiv.org/abs/2010.02658.

Bastardi, A. & Shafir, E. (1998). On the Pursuit and Misuse of Useless Information. *Journal of Personality and Social Psychology*, 75(1), 19–32.

Benkler, Y. (2006). *The Wealth of Networks: How Social Production Transforms Markets and Freedom*. New Haven: Yale University Press.

Boczkowski, P. (2021). *Abundance: On the Experience of Living in a World of Information Plenty*. Oxford: Oxford University Press.

Bodoff, D. (2024). On the Status of Machine Learning Inferences in Data Privacy Economics and Regulation. In D. R. Raban & J. Włodarczyk (Eds.), *The Elgar Companion to Information Economics* (pp. 462–480). Cheltenham, UK and Northampton, MA, USA: Edward Elgar Publishing.

Braman, S. (2006). The Micro- and Macroeconomics of Information. *Annual Review of Information Science and Technology*, 40, 3–52.

Daoud, A. (2018). Unifying Studies of Scarcity, Abundance, and Sufficiency. *Ecological Economics*, 147, 208–217.

Desai, D. R. & Lemley, M. A. (2022). Scarcity, Regulation, and the Abundance Society. *Stanford Law School Working Paper Series*, 572. https://doi.org/10.2139/ssrn.4150871.

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

Dugast, J. & Foucault, T. (2018). Data Abundance and Asset Price Informativeness. *Journal of Financial Economics*, 130(2), 367–391.
Dugger, W. M. (2009). *Economic Abundance: An Introduction*. Armonk, NY: M. E. Sharpe.
Eysenbach, G. (2020). How to Fight an Infodemic: The Four Pillars of Infodemic Management. *Journal of Medical Internet Research*, 22(6), e21820. https://doi.org/10.2196/21820.
Fox, D. M. (1967). *The Discovery of Abundance: Simon N. Patten and the Transformation of Social Theory*. Ithaca, NY: Cornell University Press.
Gino, F. & Pierce, L. (2009). The Abundance Effect: Unethical Behavior in the Presence of Wealth. *Organizational Behavior and Human Decision Processes*, 109(2), 142–155.
Global Infectious Hazard Preparedness (2021). *An Overview of Infodemic Management during COVID-19*. World Health Organization, Health Topics (May). https://www.who.int/health-topics/infodemic#tab=tab_1.
Goldfarb, A. & Tucker, C. E. (2017). *Digital Economics*. NBER Working Paper, 23684. https://ssrn.com/abstract=3023079.
Grossman, S. J. & Stiglitz, J. E. (1980). On the Impossibility of Informationally Efficient Markets. *American Economic Review*, 70(3), 393–408.
Hayek, F. A. (1945). The Use of Knowledge in Society. *American Economic Review*, 35(4), 519–530.
Heyman, J. & Ariely, D. (2004). Effort for Payment: A Tale of Two Markets. *Psychological Science*, 15(11), 787–793.
Hoeschele, W. (2010). *The Economics of Abundance: A Political Economy of Freedom, Equity, and Sustainability*. New York: Routledge.
Jefferson, T. (1905). Thomas Jefferson to Isaac McPherson. In A. A. Lipscomb & A. E. Bergh (Eds.), *The Writings of Thomas Jefferson* (Document 25). Thomas Jefferson Memorial Association.
Jones, C. I. & Tonetti, C. (2020). Nonrivalry and the Economics of Data. *American Economic Review*, 110(9), 2819–2858.
Kahneman, D. & Tversky, A. (1979). Prospect Theory: An Analysis of Decision Under Risk. *Econometrica*, 47(2), 263–291.
Konstan, J. A. & Kong, R. (2024). The Challenge of Organizational Bulk Email Systems: Models and Empirical Studies. In D. R. Raban & J. Włodarczyk (Eds.), *The Elgar Companion to Information Economics* (pp. 407–435). Cheltenham, UK and Northampton, MA, USA: Edward Elgar Publishing.
Lemley, M. A. (2016). *IP in a World Without Scarcity*. https://doi.org/10.31235/osf.io/3vy5a.
Levitan, K. B. (1982). Information Resources as "Goods" in the Life Cycle of Information Production. *Journal of the American Society for Information Science*, 33(1), 44–54.
Linde, F. & Stock, W. G. (2011). *Information Markets: A Strategic Guideline for the I-Commerce*. Berlin: De Gruyter.
Lyman, P. & Varian, H. R. (2000). *How Much Information*. http://www.sims.berkeley.edu/how-much-info.
Ma, L. (2023). Information, Platformized. *Journal of the Association for Information Science and Technology*, 74(2), 273–282.
Machiavelli, N. (1532). *The Prince*. https://www.gutenberg.org/ebooks/1232.
Madison, M. J., Frischmann, B. M., Sanfilippo, M. R., & Strandburg, K. J. (2022). Too Much of a Good Thing? A Governing Knowledge Commons Review of Abundance in Context. *Frontiers in Research Metrics and Analytics*, 7. https://doi.org/10.3389/FRMA.2022.959505.
Malthus, T. (1798). *An Essay on the Principle of Population, as it Affects the Future Improvement of Society with Remarks on the Speculations of Mr. Godwin, M. Condorcet, and Other Writers*. London: Printed for J. Johnson, in St. Paul's Church-Yard.
Mansour, E., Sambra, A. V., Hawke, S., Zereba, M., Capadisli, S., Ghanem, A., Aboulnaga, A., & Berners-Lee, T. (2016). A Demonstration of the Solid Platform for Social Web Applications. *WWW 2016 Companion – Proceedings of the 25th International Conference on World Wide Web* (pp. 223–226). https://doi.org/10.1145/2872518.2890529.
Maslow, A. H. (1943). A Theory of Human Motivation. *Psychological Review*, 50(4), 370–396.
McTaggart, J. M. E. (1896). *Studies in the Hegelian Dialectic*. Cambridge: Cambridge University Press.
Peach, J. & Dugger, W. M. (2006). An Intellectual History of Abundance. *Journal of Economic Issues*, 40(3), 693–706.

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

Plamondon, S. (2022). Inequality in Abundance. *Frontiers in Research Metrics and Analytics*, 7. https://doi.org/10.3389/FRMA.2022.980677.

Raban, D. R. (2007). User-Centered Evaluation of Information: A Research Challenge. *Internet Research*, 17(3), 306–322.

Raban, D. R. & Ahituv, N. (2024). Assessing the Perceived Value of Information in an Information Immersive World. In D. R. Raban & J. Włodarczyk (Eds.), *The Elgar Companion to Information Economics* (pp. 364–378). Cheltenham, UK and Northampton, MA, USA: Edward Elgar Publishing.

Raban, D. R., Barzilai, S., & Portnoy, L. (2019). The Unexpected Benefits of Paying for Information: The Effects of Payment on Information Source Choices and Epistemic Thinking. In M. Pańkowska & K. Sandkuhl (Eds.), *Perspectives in Business Informatics Research* (pp. 163–176). Cham: Springer.

Raban, D. R. & Koren, L. (2019). Risk as a Predictor of Online Competitive Information Acquisition. *Open Information Science*, 3(1), 47–60.

Raban, D. R. & Rafaeli, S. (2006). The Effect of Source Nature and Status on the Subjective Value of Information. *Journal of the American Society for Information Science and Technology*, 57(3), 321–329.

Rafikov, I. & Akhmetova, E. (2019). Scarcity in the Age of Abundance: Paradox and Remedies. *International Journal of Ethics and Systems*, 35(1), 119–132.

Rifkin, J. (2014). *The Zero Marginal Cost Society: The Internet Of Things, the Collaborative Commons, and the Eclipse of Capitalism*. New York: St. Martin's Press.

Robbins, L. (1932). *An Essay on the Nature and Significance of Economic Science*. London: Macmillan & Co.

Seneca, L. A. (1932). *Seneca's Letters to Lucilius*. Oxford: Clarendon Press.

Shampanier, K., Mazar, N., & Ariely, D. (2007). Zero as a Special Price: The True Value of Free Products. *Marketing Science*, 26(6), 742–757.

Shapiro, C. & Varian, H. R. (1999). *Information Rules: A Strategic Guide to the Network Economy*. Boston: Harvard Business School Press.

Sheehan, B. (2010). *The Economics of Abundance: Affluent Consumption and the Global Economy*. Cheltenham, UK and Northampton, MA, USA: Edward Elgar Publishing.

Simon, H. A. (1955). A Behavioral Model of Rational Choice. *The Quarterly Journal of Economics*, 69(1), 99–118.

Simon, H. A. (1971). Designing Organizations for an Information-Rich World. In M. Greenberger (Ed.), *Computers, Communication, and the Public Interest* (pp. 37–72). Baltimore, MD: Johns Hopkins Press.

Sims, C. A. (2003). Implications of Rational Inattention. *Journal of Monetary Economics*, 50(3), 665–690.

Spence, M. (1973). Job Market Signaling. *The Quarterly Journal of Economics*, 87(3), 355–374.

Stigler, G. J. (1961). The Economics of Information. *Journal of Political Economy*, 69(3), 213–225.

Stiglitz, J. E. (1975). The Theory of 'Screening', Education, and the Distribution of Income. *American Economic Review*, 65(3), 283–300.

Stiglitz, J. E. (2000). The Contributions of the Economics of Information to Twentieth Century Economics. *The Quarterly Journal of Economics*, 115(4), 1441–1478.

Stiglitz, J. E. (2009). Government Failure vs. Market Failure: Principles of Regulation. In E. J. Balleisen & D. A. Moss (Eds.), *Government and Markets: Toward a New Theory of Regulation* (pp. 13–51). New York: Cambridge University Press.

Stiglitz, J. E. & Kosenko, A. (2024a). Robust Theory and Fragile Practice: Information in a World of Disinformation. Part 1: Indirect Communication. In D. R. Raban & J. Włodarczyk (Eds.), *The Elgar Companion to Information Economics* (pp. 20–52). Cheltenham, UK and Northampton, MA, USA: Edward Elgar Publishing.

Stiglitz, J. E. & Kosenko, A. (2024b). Robust Theory and Fragile Practice: Information in a World of Disinformation. Part 2: Direct Communication. In D. R. Raban & J. Włodarczyk (Eds.), *The Elgar Companion to Information Economics* (pp. 53–80). Cheltenham, UK and Northampton, MA, USA: Edward Elgar Publishing.

Stock, W. G. (2024). Payment on Information Markets. In D. R. Raban & J. Włodarczyk (Eds.), *The Elgar Companion to Information Economics* (pp. 339–363). Cheltenham, UK and Northampton, MA, USA: Edward Elgar Publishing.

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access

Swan, M. (2016). Philosophy of Social Robotics: Abundance Economics. In A. Agah, J. J. Cabibihan, A. Howard, M. Salichs, & H. He (Eds.), *Social Robotics. ICSR 2016*. Lecture Notes in Computer Science, vol. 9979 (pp. 900–908). Cham: Springer.

Swan, M. (2017). Is Technological Unemployment Real? An Assessment and a Plea for Abundance Economics. In K. LaGrandeur & J. J. Hughes (Eds.), *Surviving the Machine Age: Intelligent Technology and the Transformation of Human Work* (pp. 19–33). Cham: Palgrave Macmillan.

Taleb, N. N. (2012). *Antifragile: Things That Gain from Disorder*. New York: Random House.

Thaler, R. H. (1991). The Psychology of Choice and the Assumptions of Economics. In R. H. Thaler, *Quasi Rational Economics* (pp. 137–166). New York: Russell Sage Foundation.

Tversky, A. & Kahneman, D. (1982). Judgement Under Uncertainty: Heuristics and Biases. *Science*, 185(4157), 1124–1131.

Tzu, S. (2010). *The Art of War*. Chichester: Capstone Publishing.

Waldfogel, J. (2017). How Digitization Has Created a Golden Age of Music, Movies, Books, and Television. *The Journal of Economic Perspectives*, 31(3), 195–214.

Ward, B. (1966). The Economics of Abundance. *Ekistics*, 21(123), 107–109.

Wilde, O. (1893). *Lady Windermere's Fan: A Play about a Good Woman*. London: E. Mathews and J. Lane.

Włodarczyk, J. (2020). Non solum technology – korzenie i perspektywy rozwojowe gospodarki cyfrowej. In I. Ostoj & K. Bartuś (Eds.), *Innowacje na poziomie mikro- i makroekonomicznym* (pp. 11–23). Katowice: Uniwersytet Ekonomiczny.

Włodarczyk, J. (2024). Information and Income Distribution: The Perspective of Information Economics. In D. R. Raban & J. Włodarczyk (Eds.), *The Elgar Companion to Information Economics* (pp. 339–363). Cheltenham, UK and Northampton, MA, USA: Edward Elgar Publishing.

Daphne R. Raban and Julia W ̶odarczyk - 9781802203967
Downloaded from https://www.elgaronline.com/ at 04/15/2026 05:28:46AM
via free access